

Read the passage below, making note of any important points and definitions.

After Uranus was discovered, its orbital motion was recorded over many years (orbital information was also found in historical records, where it turned out that Uranus had been seen before but mistakenly identified as a star). Newton's laws of motion were used, as usual, to calculate a predicted orbit for Uranus. Between the observed orbit and the calculated orbit, however, there was a very tiny discrepancy. Now the calculation of planetary orbits had become a highly refined and exact science by the middle of the nineteenth century. The earliest applications of Newton's laws to planetary motion assumed that the planets were influenced only by the gravitational attraction of the sun. Because the sun is many hundreds of times more massive than any planet, only considering the sun is a very good approximation. But all the planets have some mass, and so they all have some slight gravitational influence on each other. As long as the observations of the planetary positions are no more precise than the predictions of the approximate theory, these slight effects can be ignored. As the power and precision of telescopes improved, however, astronomers were able to measure the positions of the planets so precisely that the theory could no longer make accurate predictions using only the gravitational influence of the sun. The other planets had to be included.

Including the gravitational influence of the other planets is extremely difficult. Applying Newton's laws to calculate the motion of objects can only be done for (at most) two mutually interacting objects. The reason for this limitation is mathematical; the equations we get for three or more objects can't be solved. Fortunately, the great mass of the sun makes it initially possible to ignore the other planets, and only that fact made it possible for Newton to apply his theory. As I said, subsequent progress in observational astronomy eventually forced physicists to deal with this problem, which they did in a remarkably clever manner. They couldn't solve the "three body problem" directly, but since the gravitational attraction of a planet is so much smaller than that of the sun, they were able to treat it as a small correction on the orbit they could solve using only the sun. The technical jargon for this technique is that the gravity of the planet is treated as a small perturbation on the gravity of the sun. The effect of this perturbation on the orbit could be approximately calculated using special mathematical methods that were worked out for this purpose. A feature of the technique is that the approximation becomes better as the perturbation becomes smaller. For this reason, the technique works

very well for our planetary motion problem. Needless to say, the actual mathematical manipulations are exceedingly difficult. Even so, a number of brilliant mathematical physicists (Laplace being the most prominent) had used these techniques to work out all the motions of the planets by the beginning of the nineteenth century, and all of the predictions fit the observations perfectly.

Until Uranus. By the 1840s, it had become clear that there was a small discrepancy between the growing collection of observations and the increasingly precise perturbation calculations for Uranus. When I say small, keep in mind just how good the observations and calculations had become by that time. The discrepancy we are discussing is roughly 30 seconds of arc, about the same as the angle between two automobile headlights which you see from five miles away. But as small as the discrepancy was, it was still real and it shouldn't have been there. How could this discrepancy be explained? One "obvious" explanation is that the inverse-distance-squared relation for gravity breaks down in the case of Uranus and some different law holds (perhaps due to its much greater distance from the sun). Such a revolutionary and arbitrary conclusion is unwarranted by the facts of this case, however; throwing out universal gravitation should be our last resort, not our first choice. Another explanation might be that a large body (e.g., a comet) had passed close to Uranus sometime in the previous hundred years, greatly perturbing its orbit. This hypothesis is plausible, but it has the disadvantage of being untestable. Other possible explanations might be invented, but the most popular proposed explanation at the time is also the one we are most interested in here: Another planet exists in our solar system, exerting a gravitational perturbation that accounts for the discrepancies in the orbit of Uranus.

If you make the hypothesis that another planet, yet unknown, is the cause of the irregularities in the orbit of Uranus, then your next step would naturally be an attempt to locate the new planet. In particular, you would want to calculate the orbit that this new planet must have in order to account, quantitatively, for these irregularities. The difficulty of such a calculation is mind-boggling. It's hard enough to calculate the perturbations on an orbit from a source that you know, but here you must work backwards and figure out the location of the source based on the perturbations. Yet two formidable mathematicians, J. C. Adams in England and U. LeVerrier in France, undertook this calculation and they both came to the same conclusion. The world now had a prediction of where to look for the hypothetical new planet. LeVerrier sent his work to the observational

astronomer J. G. Galle in 1846. Galle immediately set to work looking for the planet. He found it on his first night of observation! The new planet eventually came to be named Neptune. This exciting discovery was a great triumph for Newton's mechanics.

Gregory N Derry. What science is and how it works. Princeton University Press, 1999.

Read the passage below, making note of any important points and definitions.

Edward Jenner started his career as an apprentice country doctor in Gloucestershire, the rural area of England where he grew up. After his apprenticeship, Jenner went to London in 1770 for more advanced training under the highly regarded surgeon, John Hunter. Medicine was still in a somewhat primitive state at this time, using many traditional methods of doubtful efficacy. Hunter was a pioneer in the application of scientific thinking to medical practice, and he taught Jenner to do the same. Jenner proved to be an excellent student, and he developed into a first rate doctor under the guidance of the brilliant Hunter. Equally able as both a medical practitioner and as a scientist, Jenner embarked on his own career after his time with Hunter ended.

Against his teacher's wishes, Jenner decided to move back to Gloucestershire (Hunter wanted Jenner to stay in London, where he could make a reputation). His move back to the countryside, however, gave Jenner the opportunity to follow up on an idea he had gotten when he was still a young apprentice. While treating a milkmaid for a minor ailment known as cowpox, Jenner had become acquainted with one of the local legends of the Gloucestershire region. The milkmaid told him that she was lucky to have the cowpox, because now she would never contract smallpox. Smallpox was one of the most dreaded diseases of the time, and the milkmaid assured Jenner that having had cowpox protects against getting smallpox. Jenner put this conversation in the back of his mind at the time, but now he was hoping to look into the matter more thoroughly. Gloucestershire was a major dairy farming area, and this made it an ideal place to conduct his study. Cowpox was a disease that the cows contracted, and the cows often then transmitted the illness to humans (through cuts on their hands, for example) as they milked the cows. Cowpox wasn't very serious; it caused fever, aching, and some temporary blisters around the hands. The illness lasted a few days, and a full recovery could be expected. The

cowpox sometimes came to the dairies in epidemics, but sometimes it vanished for years on end.

Smallpox at that time was a worldwide scourge, highly contagious and often fatal. In the century before Jenner began his work, smallpox had claimed over twenty million people in Europe. Almost a third of the children under age three in Britain succumbed to the Red Death. The smallpox sometimes raged unchecked in terrible epidemics, and there was no treatment available. Among the victims who did not die, many were left horribly disfigured, blinded, or insane. The only preventative measure known was inoculation, the practice of purposely infecting people with material from active smallpox pustules. This action might produce a less severe case of the illness, which then protects the person against contracting it again. But, the procedure often could go awry and produce a severe case, even death. Worse yet, even when the procedure worked well, the inoculated person might give the disease to others. In Russia, an entire epidemic had started this way. So Jenner's idea that there might be a safe way to prevent smallpox was exciting. It had taken root in his mind and become his dream: a world free from the Red Death. But the matter was not simple. There were a number of cases in which people who had once had cowpox did come down with the smallpox. For that reason, many of the local Gloucestershire doctors dismissed the old legend completely. And yet, there was enough anecdotal evidence in favor of the legend to still convince many people of its truth. Jenner, excellent scientist that he was, realized that he needed to start making careful observations, including keeping good notes and records, if he wished to untangle the situation.

After his medical practice was established, Jenner began his work in earnest. He made a scientific study of the cowpox, which no one had ever done before. A more precise description of the symptoms and course of the disease was needed, both in cows and in humans. For several years, Jenner carefully observed all the cases which occurred in the dairies, and he interviewed people who had gotten cowpox in the past. Making careful notes on these case histories, he began to achieve a more thorough understanding of the cowpox. At the same time, Jenner began to make a systematic study of the cases in which cowpox had apparently conferred immunity to smallpox. Just as importantly, he also studied those cases where it had not done so. If he wished to use cowpox as a tool in the fight against smallpox, Jenner would have to solve the puzzle of why some people still contract smallpox even after having cowpox. These cases were often taken to be proof that the old legend was merely superstition, and Jenner's

work was widely regarded by his colleagues as a waste of time.

But Jenner did not give up easily, and he continued to look for some clue that would solve this puzzle. As he pored over his records, he came to realize that different sets of symptoms were observed (the appearance of the pustules, swelling in the armpits, headaches, body pains, vomiting, etc.) in different victims; in other words, the cowpox had no single fixed description. The same thing was true in cows (sometimes the pustules were circular, sometimes irregular; sometimes they lasted weeks, sometimes days, and so on). Jenner concluded that what dairy farmers had been calling cowpox was actually several distinctly different diseases. This fact solved Jenner's puzzle, because only one of these diseases conferred immunity to the smallpox. Once he had this idea with which to organize his observations, Jenner was soon able to distinguish these different versions from each other. His next step was to determine which disease (he referred to it as the true cowpox) was able to protect against smallpox. Based on his records and observations, Jenner was able to give a very complete and accurate description of the true cowpox. One of the major clues that helped him was the lack of symptoms in response to inoculation with smallpox matter on the part of people who had contracted the true cowpox in the past. In this manner, after five years of patient work, Jenner was able to distill a hypothesis from his observations.

He then put his hypothesis to the test, and he discovered that some mysteries still remained. At one of the local, there was a major outbreak of the true cowpox. Jenner continued to keep his meticulous records, and so there was no doubt in his mind when these same milkers came down with smallpox the following year. His hypothesis, and his dream of defeating smallpox, seemed shattered. He pored over his records and continued to study the cowpox, looking for a solution to this new puzzle. For several more years, Jenner tried in vain to figure out why even the true cowpox sometimes failed to protect against smallpox. There seemed to be no answer. Then, while looking at two cows in different stages of the disease, Jenner realized the factor he had been overlooking for so long. The disease, and in particular the appearance of the pustules, gains in strength for a few days, then the disease is at its worst for a while, and finally it declines and goes away over a few days. This much Jenner had known well for years. But now he hypothesized that the virulence of the matter in the pustules, which transmits the disease, should also likewise gain and decline in strength; and cowpox only protects against smallpox when the matter in the pustules is at its strongest. This new hypothesis solved

the puzzle, and was consistent with the facts he knew. For example, the milkers whose smallpox epidemic so mystified him had gotten cowpox in its earliest stages.

Jenner now designed an experiment to test this latest hypothesis. In May of 1796, Jenner extracted some material from a pustule on the hands of a milkmaid named Sarah Nelmes. She had contracted the disease from a cow while it was at its worst, and her own case was also now at its strongest. These conditions were ideal for Jenner's experiment, and he used the material to purposely infect a young boy named James Phipps. After the cowpox had run its course in young Phipps, Jenner inoculated him with live smallpox matter the following July. Tensely, day after day, Jenner and the Phipps family looked for any sign of a smallpox infection beginning. But even several days after the expected time of onset, the boy had absolutely no smallpox symptoms! The experiment had succeeded. The cowpox material, deliberately introduced into a human body, had been shown to confer immunity to the Red Death.

Our story ends here, but Edward Jenner's story went on for several more years. He had an uphill battle convincing the medical community and the general public that his method, which came to be known as vaccination (from the Latin word for cow), was an effective method to prevent smallpox. His problems were compounded by incompetent people who tried to steal his idea but couldn't perform the procedures properly (in one terrible case, a quack mixed up cowpox and smallpox material, actually starting an epidemic). Such mishaps gave the vaccine an undeserved reputation for being unsafe and ineffectual. But Jenner managed to sort these problems out, and in the end his smallpox vaccine derived from the cowpox material came into widespread use, saving untold numbers of people from the ravages of smallpox. For this service to humanity, Jenner became a hero in his own time and had numerous honors bestowed on him.

There are certainly elements of luck and inspiration in this story, but it mainly illustrates the pathway to discovery that we now often call the hypothetico-deductive method. We start by making observations; organize these observations into a hypothesis; test the hypothesis against further observations and modify it as needed; make predictions based on the modified hypothesis and design experiments to test our predictions. This highly successful methodology is also sometimes enshrined in elementary textbooks as the "scientific method." The discovery of the smallpox vaccine is a good example of just how powerful this method can be.

Gregory N Derry. What science is and how it works.
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